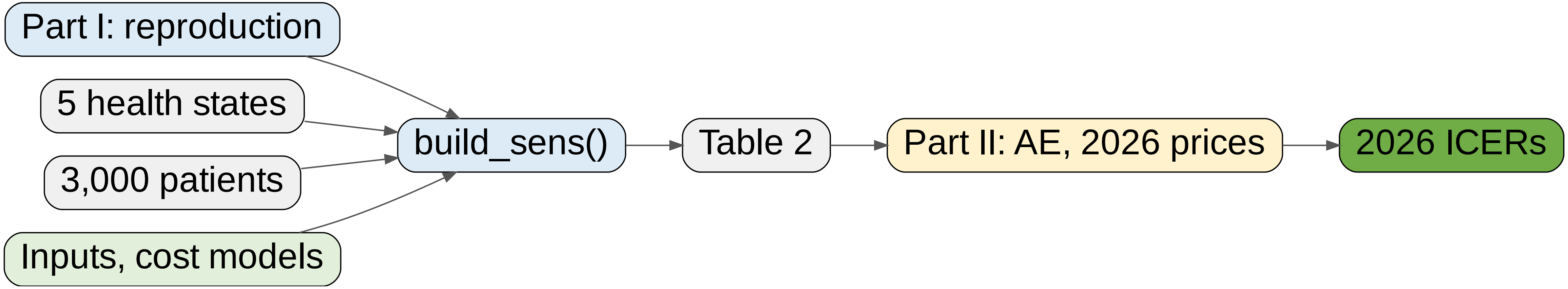
Cost-effectiveness of Sodium-Glucose Co-Transporter 2 Inhibitors (SGLT2i) in Heart Failure with Preserved Ejection Fraction (HFpEF)

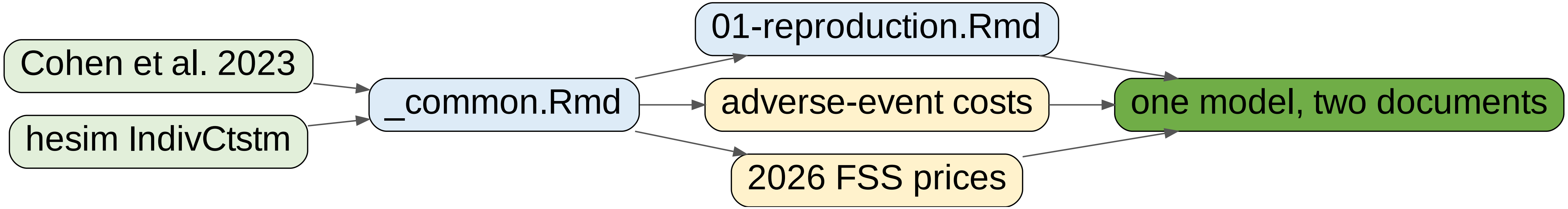
A `hesim` re-implementation of Cohen et al. (2023), extended to adverse-event costs and 2026 prices

Table of contents



What we cover

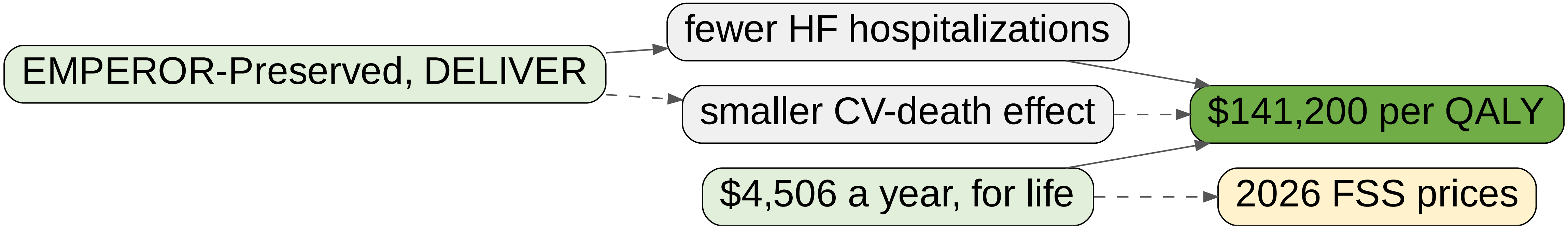
Cohen et al. (2023)[1] re-implemented on `hesim`,[2] then extended.



model definition · adverse events · 2026 prices.

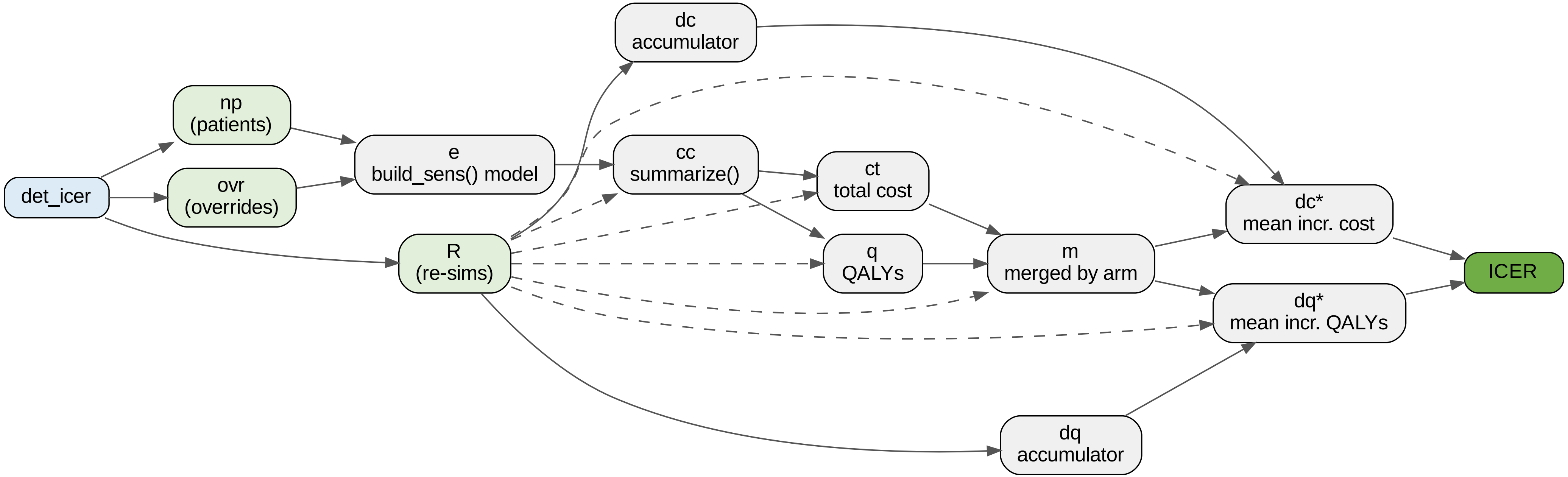
The clinical and economic question

$141,200 per QALY at the 2022 FSS price of $4,506[1,6]: intermediate value, just under $150,000.



trial inputs · the $4,506 price · the reproduced headline.

Part I: how the variables flow



`det_icer()`: the deterministic ICER behind Figure 3.

e = build_sens() · cc = summarize() · ICER.

Model structure

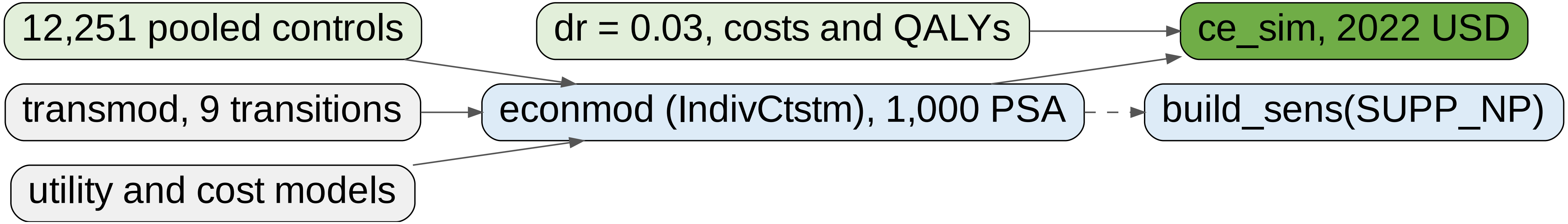
Five health states, nine transitions, individual-level and continuous-time.

From	To	Transitions
Stable	HF hosp., CV death, non-CV death	1, 2, 3
HF hosp.	Post-HF hosp., CV death, non-CV death	4, 5, 6
Post-HF hosp.	HF rehosp., CV death, non-CV death	7, 8, 9
CV / non-CV death	absorbing	–

Separate CV and non-CV mortality streams,[7,8] decaying rehospitalization risk.[9-14]

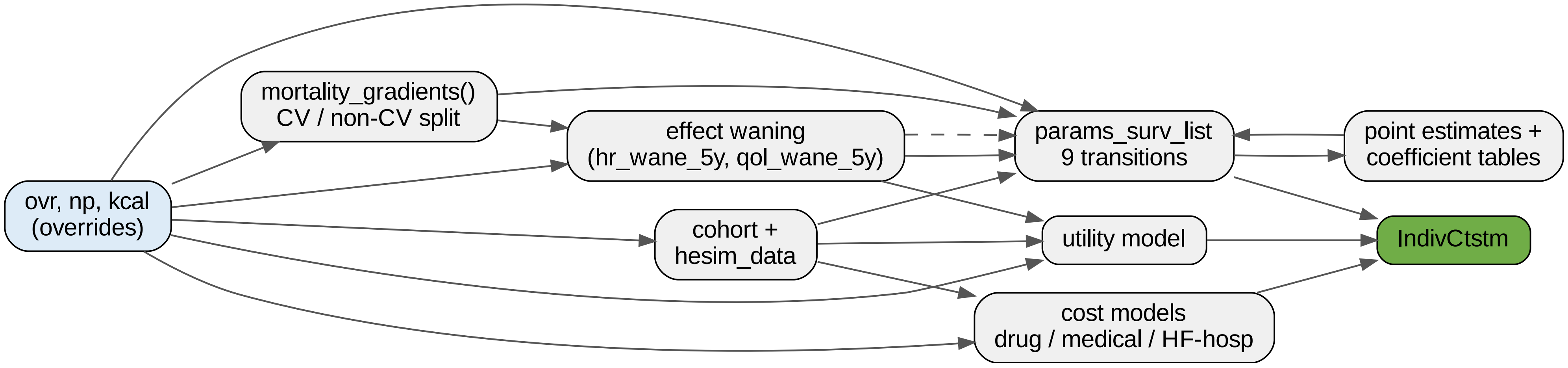
Simulation set-up

`hesim`[2] continuous time; pooled trial controls;[3,4] Second Panel case.[15]



nine transitions · cost models · deterministic runs.

Inside `build_sens()`



All 67 variables, collapsed onto the function's own sections.

nine transitions · cost models · overrides · IndivCtstm.

Key inputs

Parameter	Base case	Range
First HF hospitalization rate, /100 py[3,4]	6.26	5.89–6.71
HR for HF hospitalization vs SoC[5]	0.74	0.67–0.83
In-trial CV death, /100 py[3,4]	3.56	3.25–3.89
HR for cardiovascular death vs SoC[5]	0.88	0.77–1.00
In-trial all-cause mortality, /100 py[3,4]	7.16	6.72–7.63
Probability of discontinuing SGLT2i, %[3,4]	18.62	17.66–19.70
Baseline utility (KCCQ to EQ-5D)[4,16,17]	0.7830	0.7802–0.7859
Utility gain on SGLT2i at 12 mo[16,17]	0.0084	0.0040–0.0128

Costs

Cost item	Base case	Basis
SGLT2i, per year (2022 anchor)	$4,506	FSS Big-4, pooled[6]
net of 18.62% discontinuation	$3,667	Stable and Post-HF only
Background medical, per year	$23,004–$32,998	Three bands, model y 0/3/13[18-20]
HF hospitalization, per admission	$12,528	HCUPnet NIS[21] + prof. fee[22]

- Utility: –0.0007/y age decrement,[23] –0.16 while hospitalized.[18,24] Drug cost accrues in on-treatment living states only.

Calibration against the trial period (eTable 6)

Per 100 person-years; every HR reproduces, HF hosp. runs 8% low.

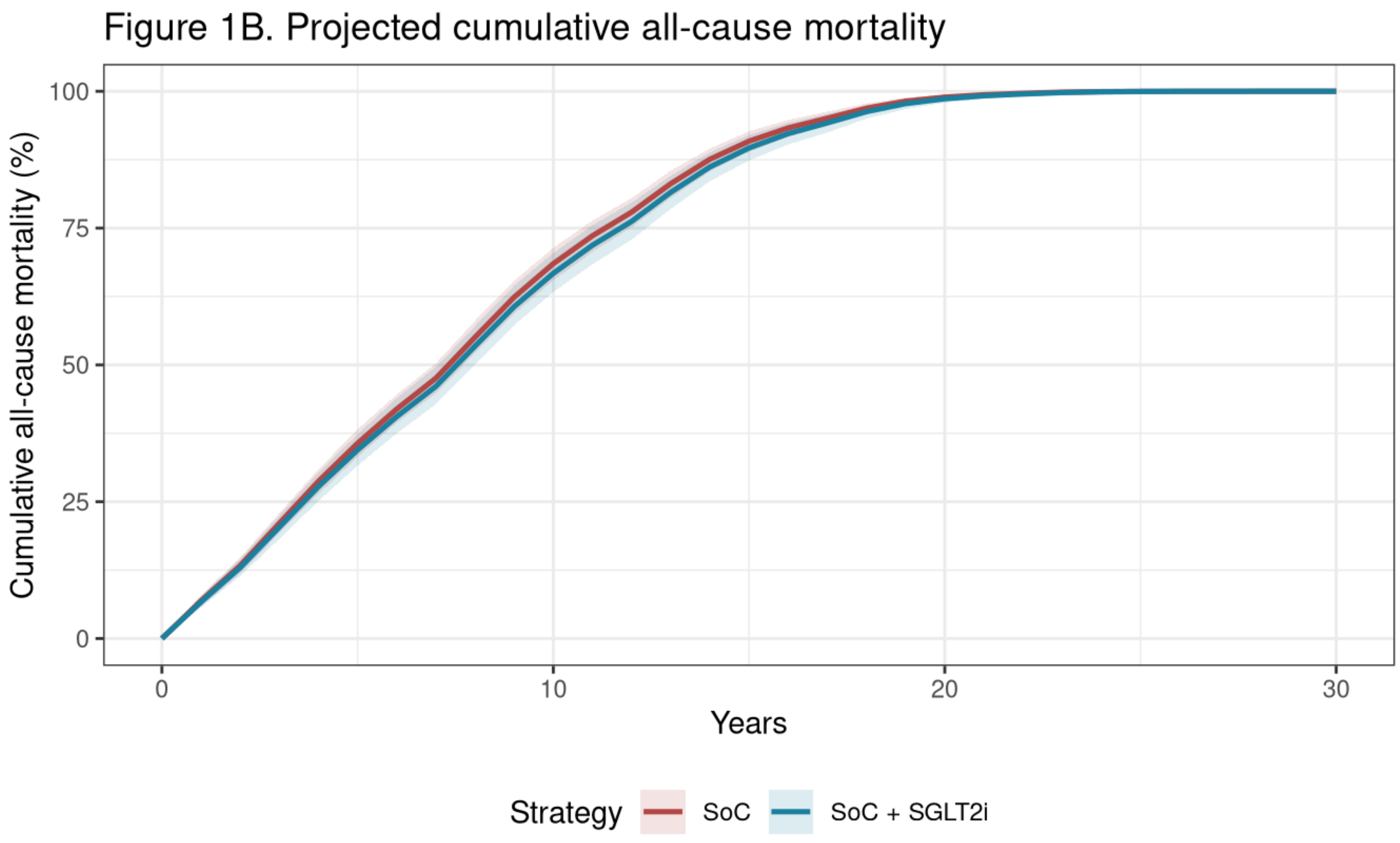
Metric	Validation target	Simulated
First HF hospitalization, SoC	6.3 (5.8–6.7)	5.8 (5.2–6.5)
First HF hospitalization, +SGLT2i	4.7 (4.3–5.0)	4.4 (3.7–5.1)
HF hospitalization hazard ratio	0.74 (0.67–0.83)	0.76 (0.63–0.89)
Cardiovascular death, SoC	3.6 (3.2–3.9)	3.6 (3.0–4.2)
Cardiovascular death hazard ratio	0.88 (0.77–1.00)	0.89 (0.70–1.10)
All-cause death, SoC	7.2 (6.7–7.6)	7.2 (6.3–8.0)
All-cause death hazard ratio	0.97 (0.88–1.06)	0.97 (0.84–1.13)

Headline validation: Table 2

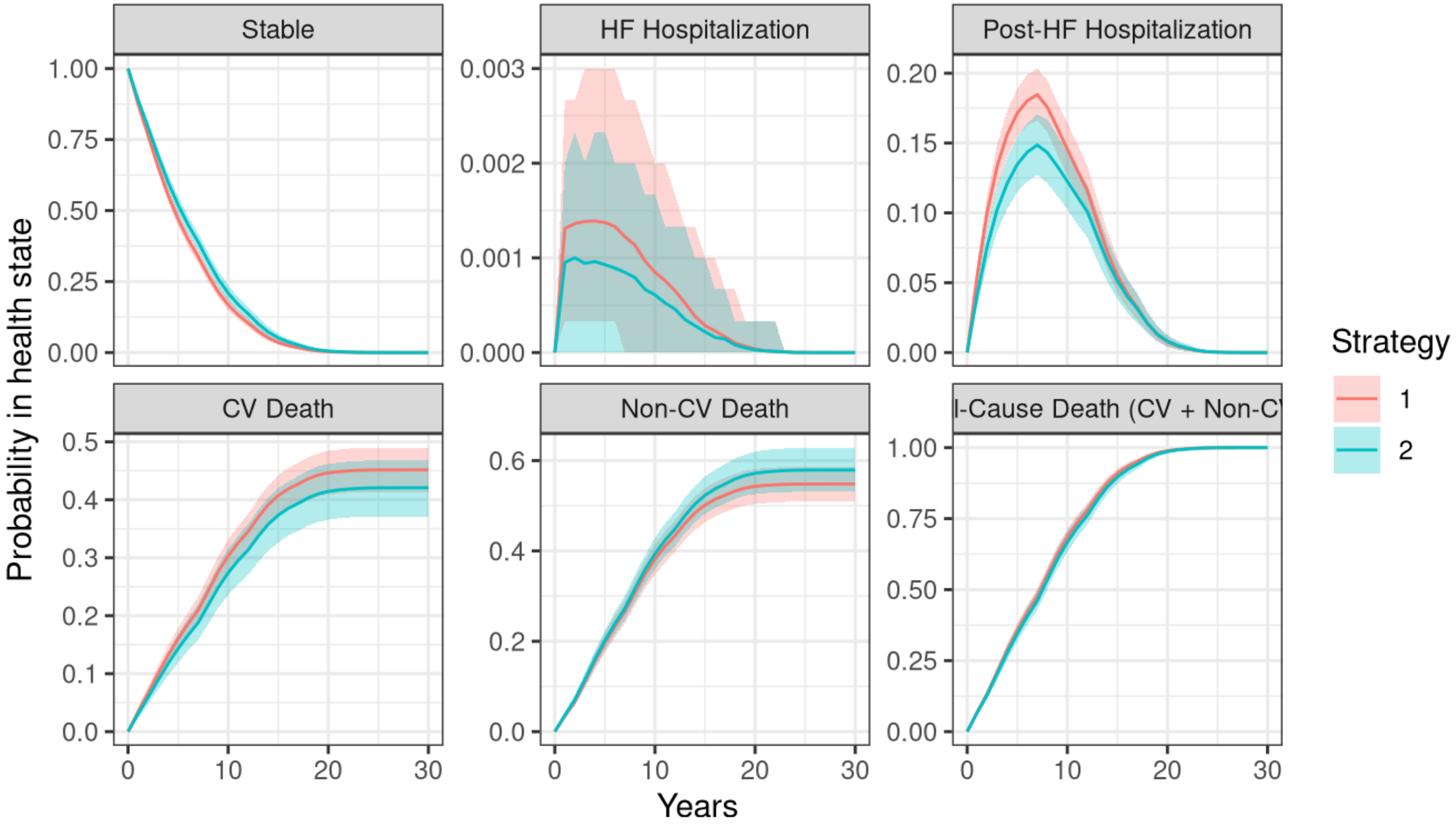
$4,506, 1,000 PSA means. Preferred at $150k: 53.1% (paper 59.1%).

Outcome	SoC	SoC + SGLT2i	Incremental	Cohen 2023
HF hospitalizations	0.85	0.59	–0.26	–0.22
Survival, undiscounted (y)	7.75	7.97	+0.22	+0.22
Life-years, discounted	6.62	6.78	+0.16	+0.16
QALYs, discounted	5.16	5.34	+0.18	+0.19
Total lifetime cost	$167,367	$193,765	$26,398	$26,312
ICER ($/QALY)			**$144,612**	**$141,200**

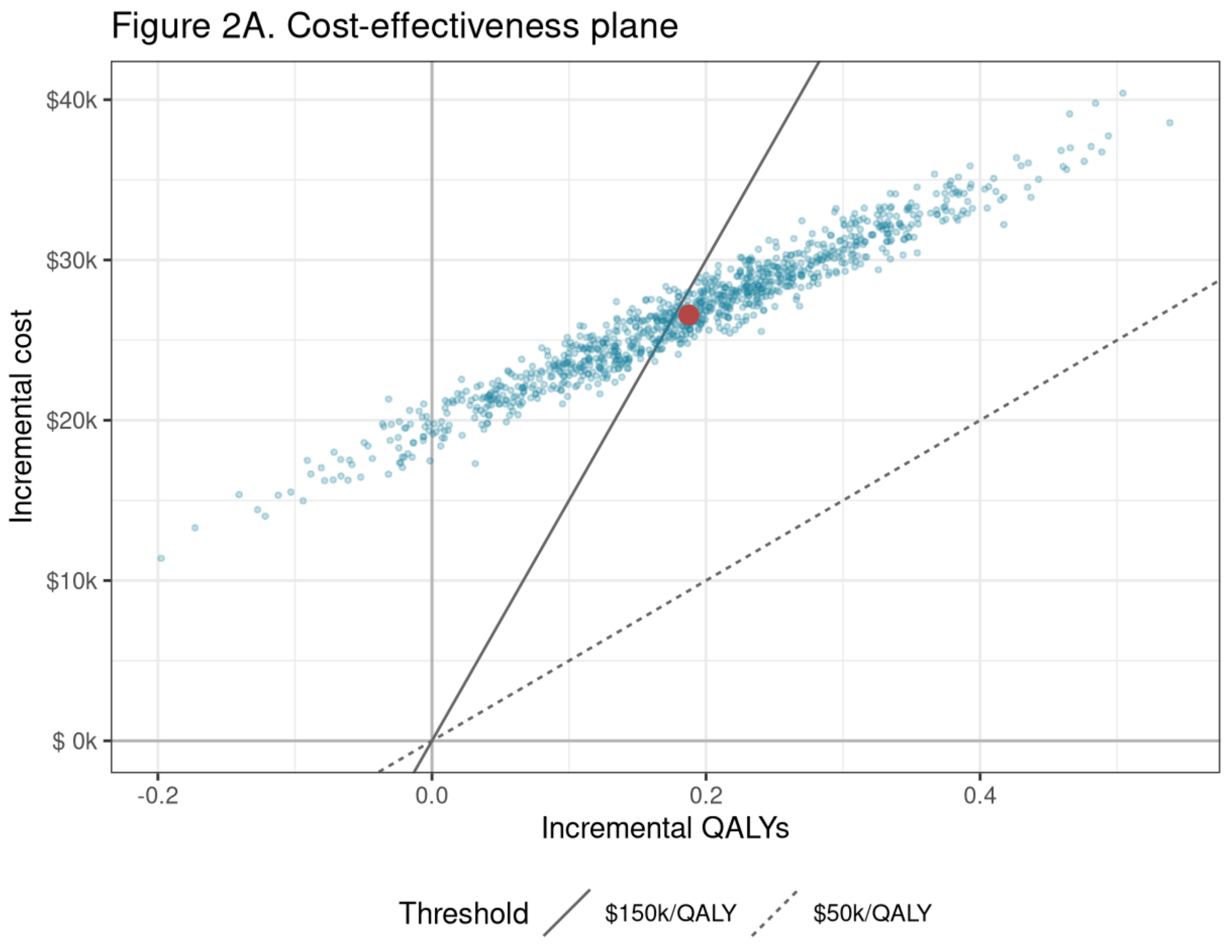
Projected all-cause mortality (Figure 1B)



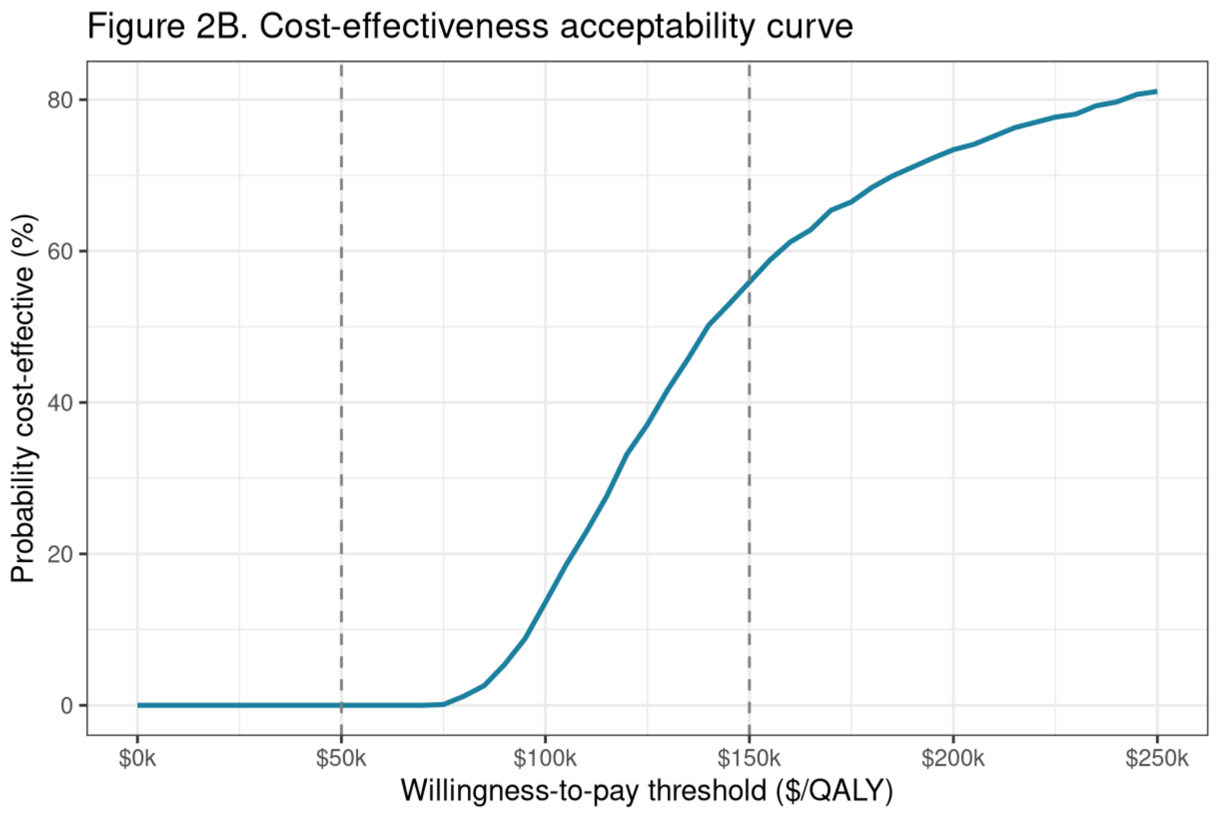
State occupancy over 30 years



Cost-effectiveness plane (Figure 2A)

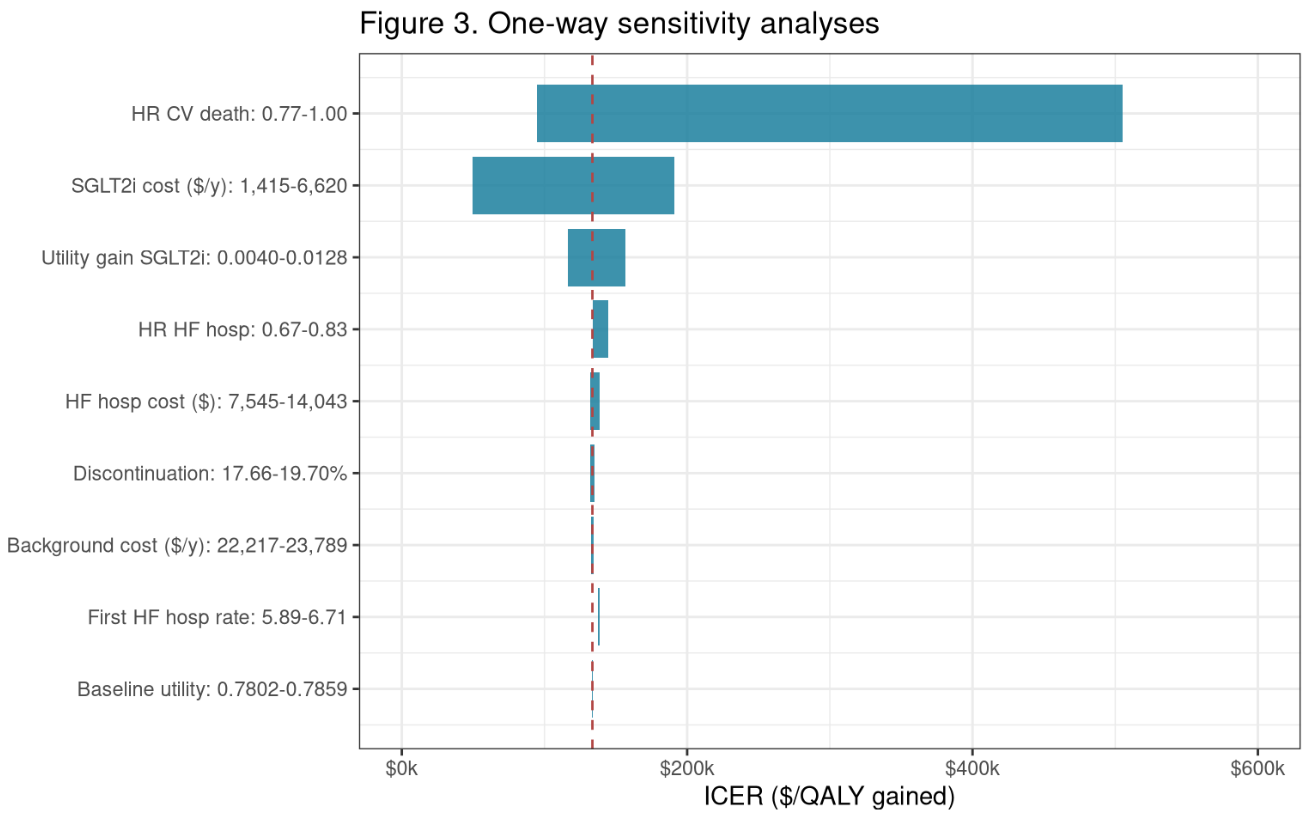


Acceptability curve (Figure 2B)



Probability preferred at $150,000/QALY: **53.1%** (paper: 59.1%).

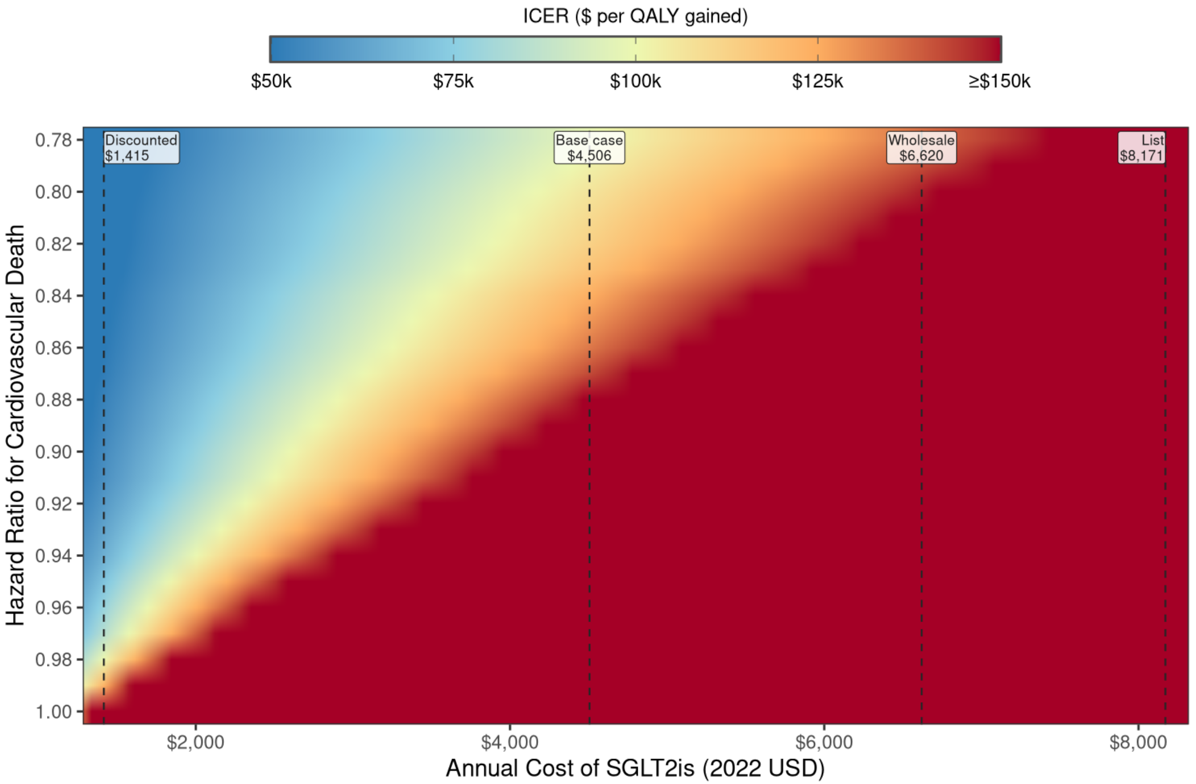
One-way sensitivity analysis (Figure 3)



Deterministic ICER **$133,584/QALY**; CV-death HR dominates, **$505k/QALY** at HR 1.00.

Two-way: price against CV-mortality effect (eFigure 4)

eFigure 4. Two-way Sensitivity Analyses When Simultaneously Varying SGLT2i Cost and Effect on Cardiovascular Mortality Rate



ICER indicates incremental cost-effectiveness ratio; QALY, quality-adjusted life-year; SGLT2i, sodium-glucose cotransporter-2 inhibitor.

Scenario analyses (eTable 8)

Deterministic point estimates, 3,500 patients per scenario.

Scenario	Incr. QALYs	ICER	Paper
Base case	0.18	\$146,686	\$141,200
Post-trial non-CV death persists	0.12	\$202,814	–
Post-trial HRs wane to 0 by 5 y	0.13	\$196,254	\$221,400
MEPS baseline QoL (0.708) + benefit	0.17	\$156,979	–
MEPS baseline QoL, no benefit	0.11	\$237,750	–
QoL benefit wanes by 5 y	0.16	\$168,925	\$163,500
EMPEROR-Preserved specific	0.17	\$135,826	–
DELIVER specific	0.19	\$158,736	–

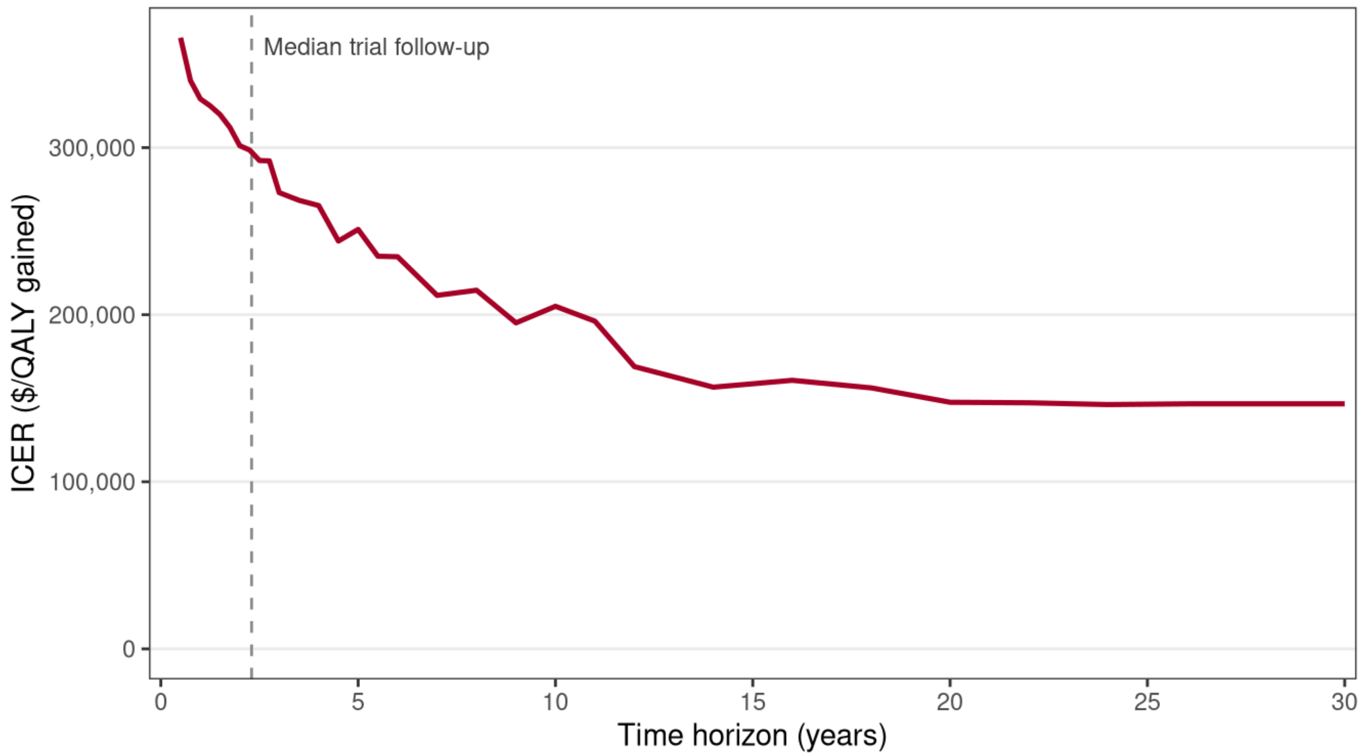
If SGLT2i had no mortality effect (eTables 9–10)

Scenario	Incr. QALYs	ICER	Paper
No mortality effect, base case	0.04	$477,002	$373,400
+ HRs wane by 5 y	0.07	$352,056	$410,800
+ MEPS QoL, with benefit	0.05	$464,645	$374,600
+ MEPS QoL, no benefit	–0.01	dominated	$4,359,500
+ QoL benefit wanes by 5 y	0.02	$973,658	$649,000

With all mortality HRs at 1, incremental survival is ≈ 0; the benefit is quality of life plus fewer hospitalizations. Every ICER lands far above $150k.

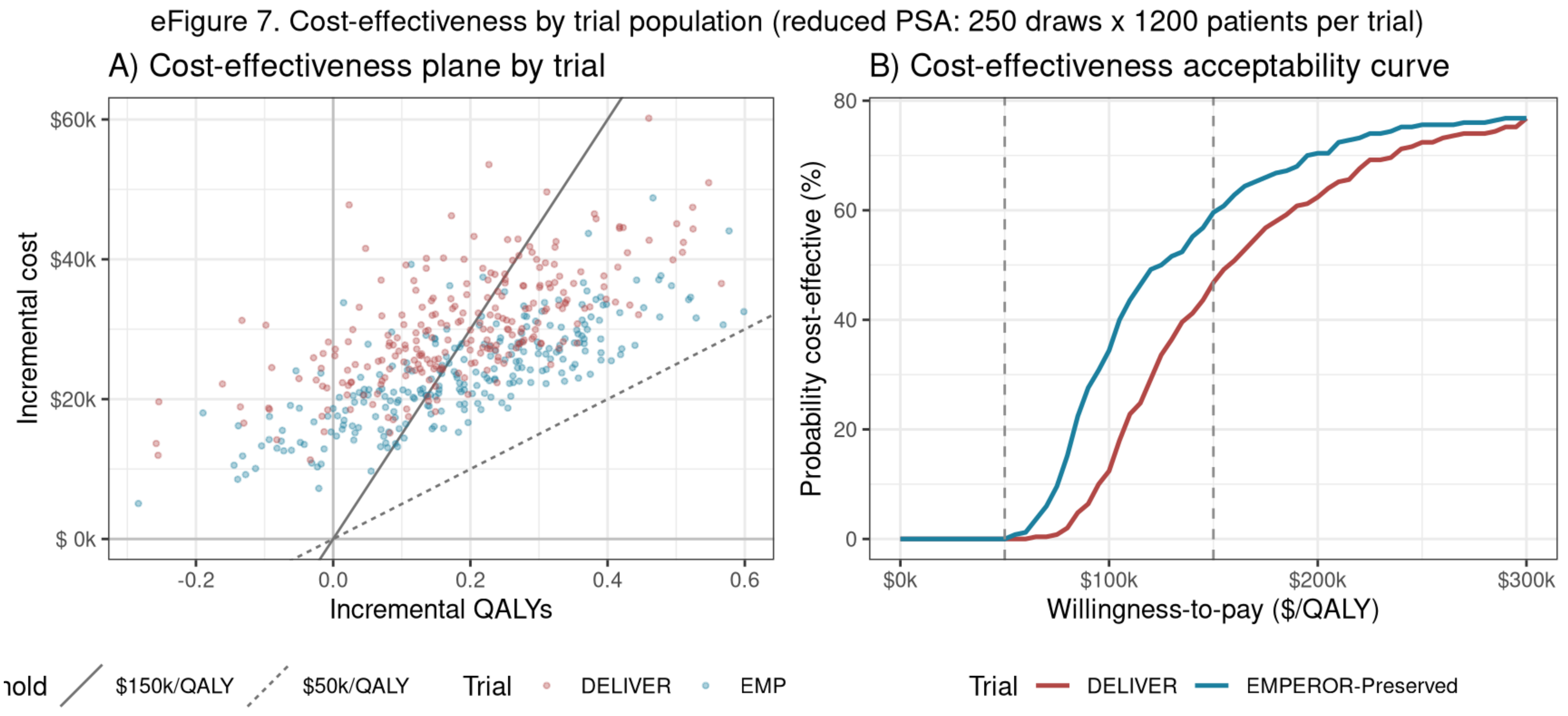
Cost-effectiveness by time horizon (eFigure 6)

eFigure 6. Cost-effectiveness when varying the time horizon

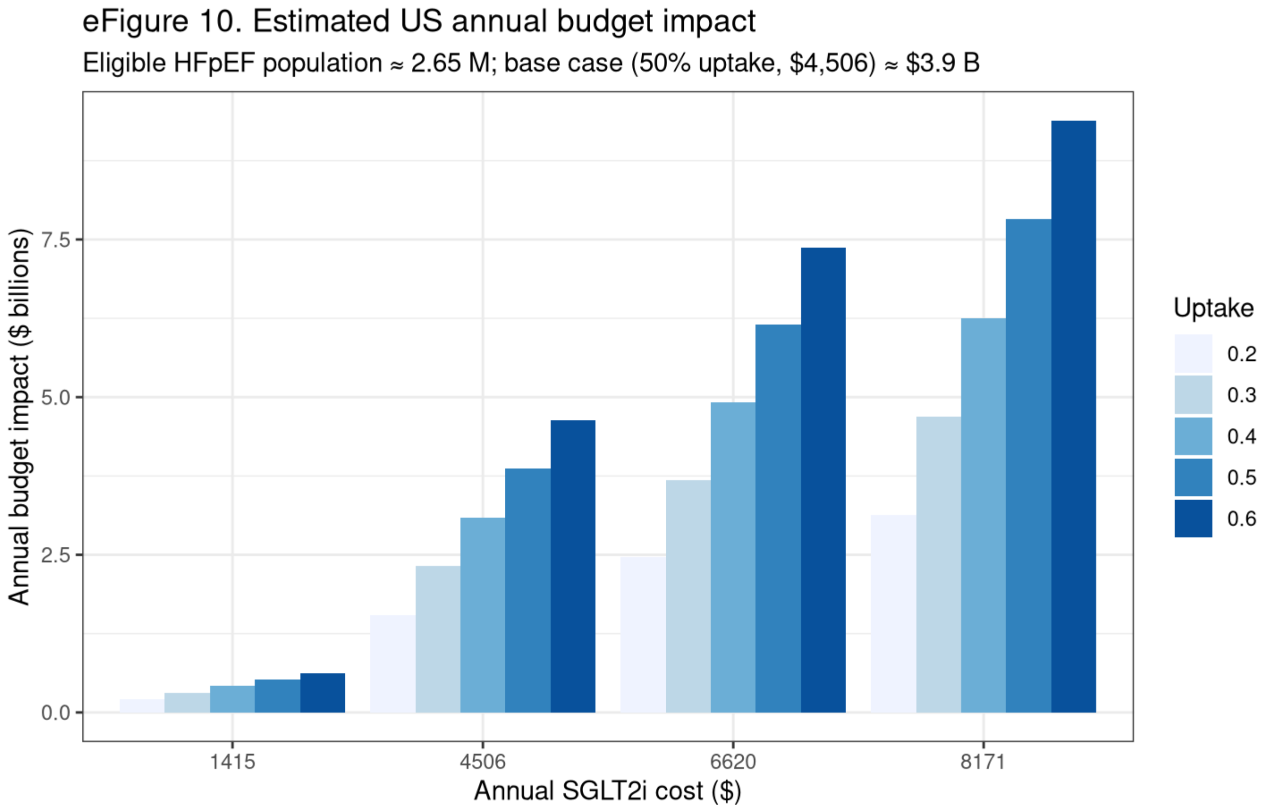


Converges to **$146,686/QALY** by year 30; short horizons unfavourable.

Cost-effectiveness by trial population (eFigure 7)



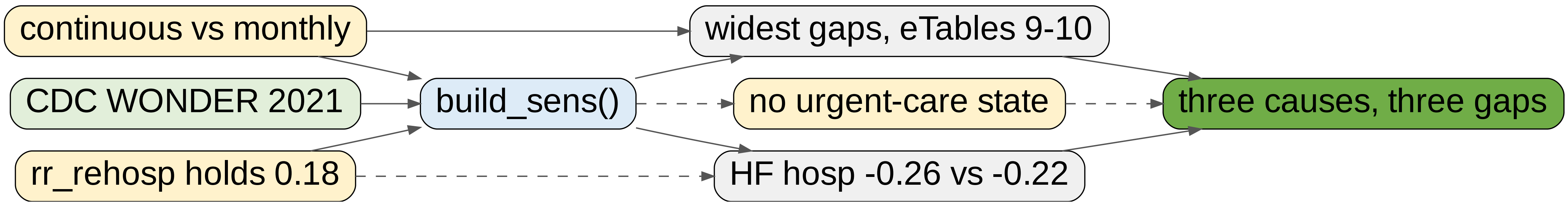
US budget impact (eFigure 10)



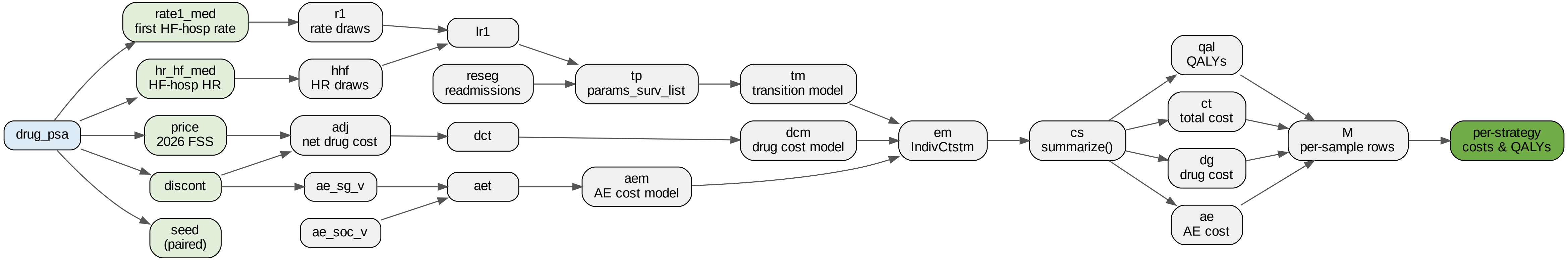
Eligible population $\approx$ **2.65 M**; base case (50% uptake at $4,506) $\approx$ **$3.9 B/y** against the published $4.3 B.

Where this reproduction diverges, and why

CDC WONDER 2021 cause split,[8] no urgent-care state,[25] frozen readmission risk.[14]



Part II: how the variables flow

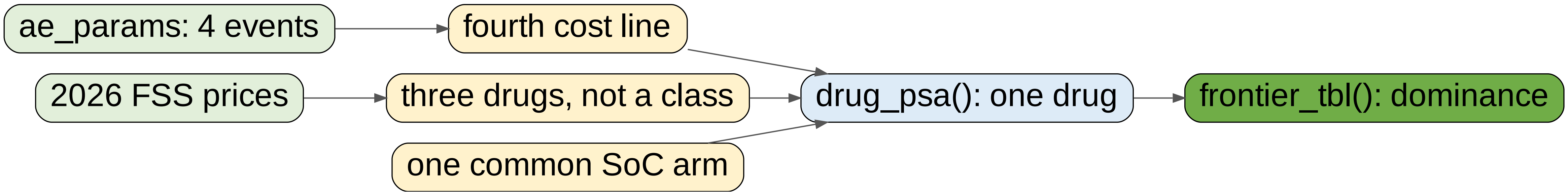


`drug_psa()`: one drug against the common SoC arm.

`tp` = nine transitions · `aem` = adverse-event costs · `adj` = 2026 net price · result.

What the extension adds

Everything so far validates the model. Cohen et al. answer neither new question.[1]



AE inputs · 2026 FSS prices · the frontier.

Adverse-event inputs

Per person-year, 2022 USD.

Event	SoC	SGLT2i	Unit cost
Genital mycotic infection (uncomplicated)[26]	0.00100	0.00600	$130
Cystitis / lower UTI (uncomplicated)[26]	0.04100	0.05100	$200
Complicated cystitis / pyelonephritis[26]	0.00200	0.00300	$11,000
Diabetic ketoacidosis[3,4]	0.00036	0.00044	$17,000

SoC $36.53 vs SGLT2i $51.41: excess **$14.88**/py.

Adverse events: the answer is “immaterial”

Drug	Incr. AE cost	% of incr. cost	ICER	ICER, no AE
Empagliflozin	$86	0.41%	$114,271	$113,799
Dapagliflozin (brand)	$96	0.29%	$178,847	$178,329
Dapagliflozin (generic)	$96	0.51%	$102,367	$101,848

- Complicated genitourinary infection supplies **74%** of the excess, ketoacidosis **8.3%**.
- Largest ICER movement **$518**/**QALY**; the frontier is unchanged.

Three drugs at 2026 FSS Big-4 prices

Drug	Product	2026 price/y	Trial	HF HR
Empagliflozin	JARDIANCE	$3,761	EMPEROR-Preserved[3]	0.71
Dapagliflozin (brand)	FARXIGA	$5,340	DELIVER[4]	0.77
Dapagliflozin (generic)	Prasco	$2,898	DELIVER[4]	0.77

10 mg, 30-count, monthly × 12; VA file of 2026-03-01.[27] Big-4, the same basis as the 2022 anchor.[6]

Headline results: three drugs as equals

1,000 PSA samples, 3,000 patients, lifetime, 3% discounting.

Drug	Incr. cost	Incr. QALYs	ICER	Preferred @ $150k
Dapagliflozin (generic)	$18,979	0.185	**$102,367**	75.1%
Empagliflozin	$20,929	0.183	**$114,271**	70.8%
Dapagliflozin (brand)	$33,159	0.185	**$178,847**	34.3%

At 2026 prices generic dapagliflozin and empagliflozin fall below $150,000/QALY; brand does not.[28]

Figure 4A–4B: ICER by drug and the CE plane

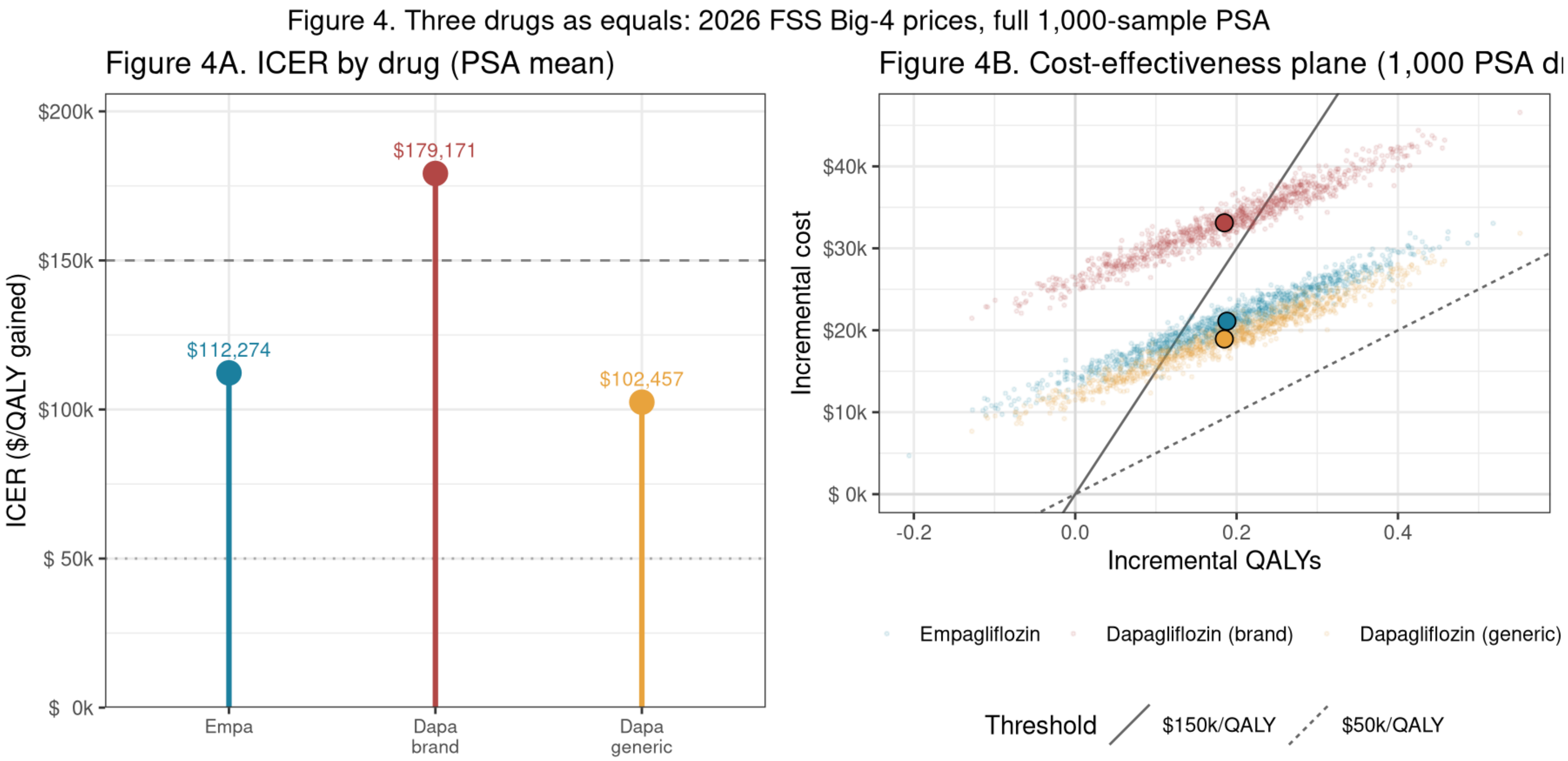


Figure 4C: Acceptability curves

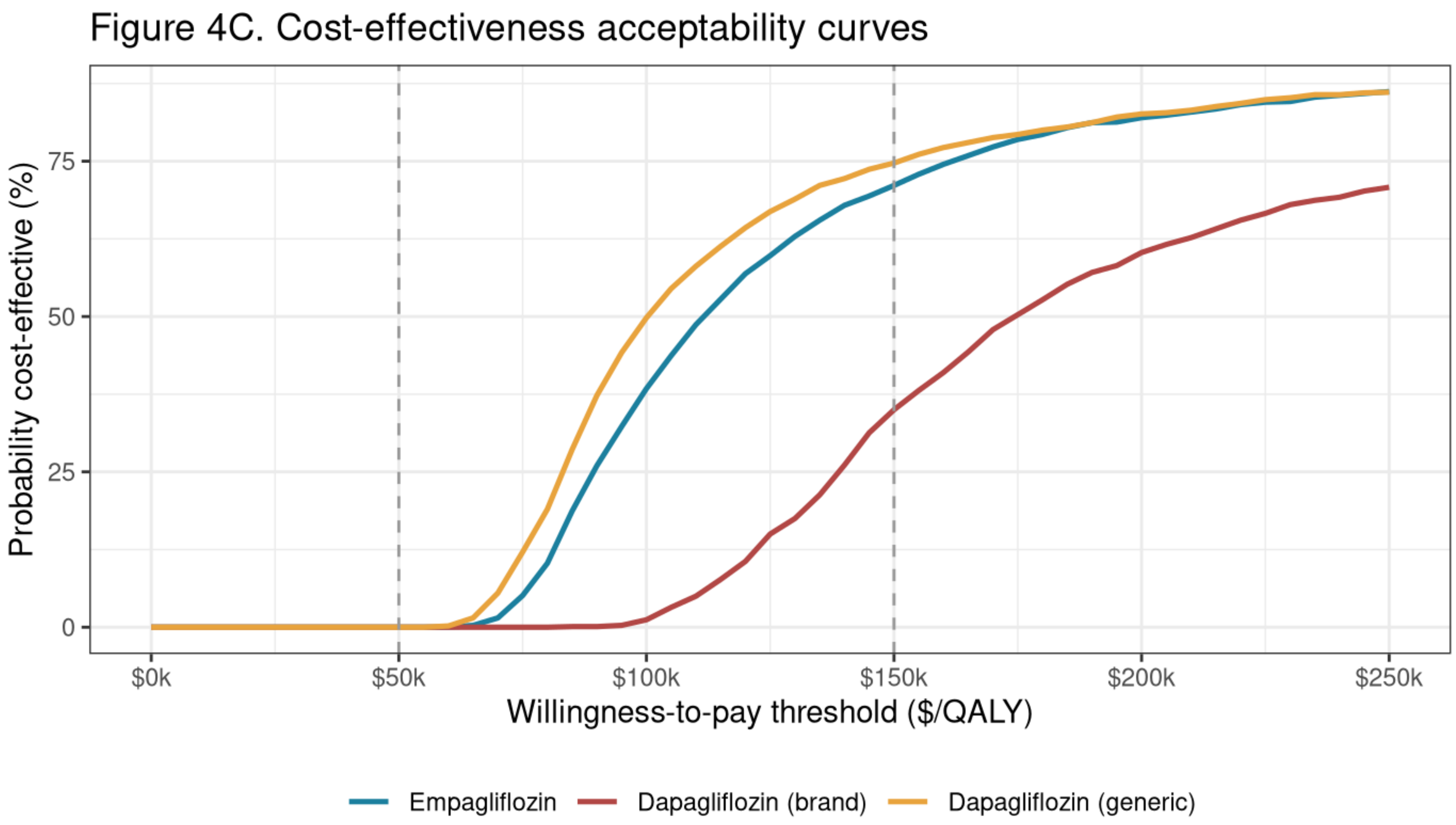


Figure 4D: Acceptability frontier

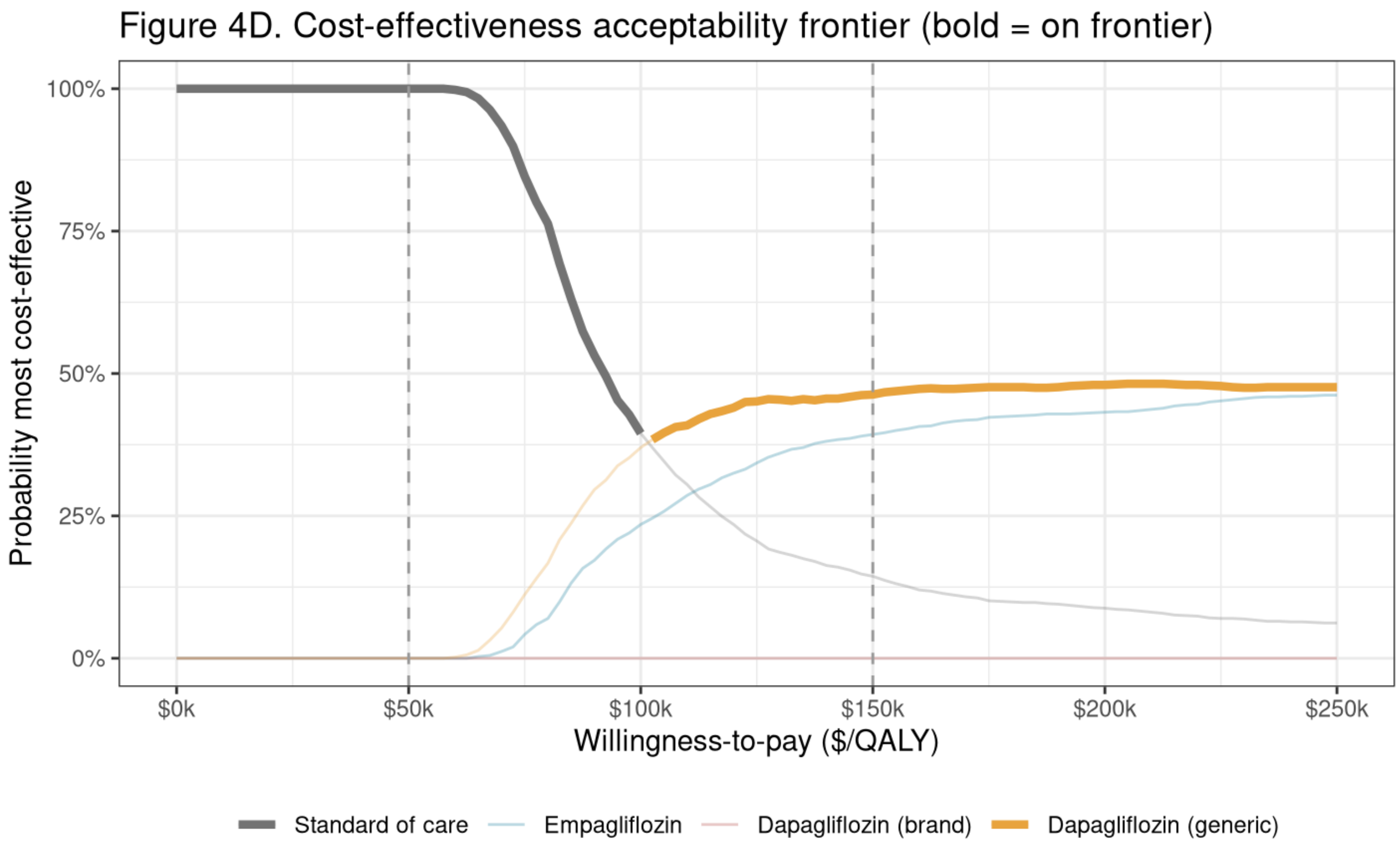


Figure 4E: Plane against the common comparator

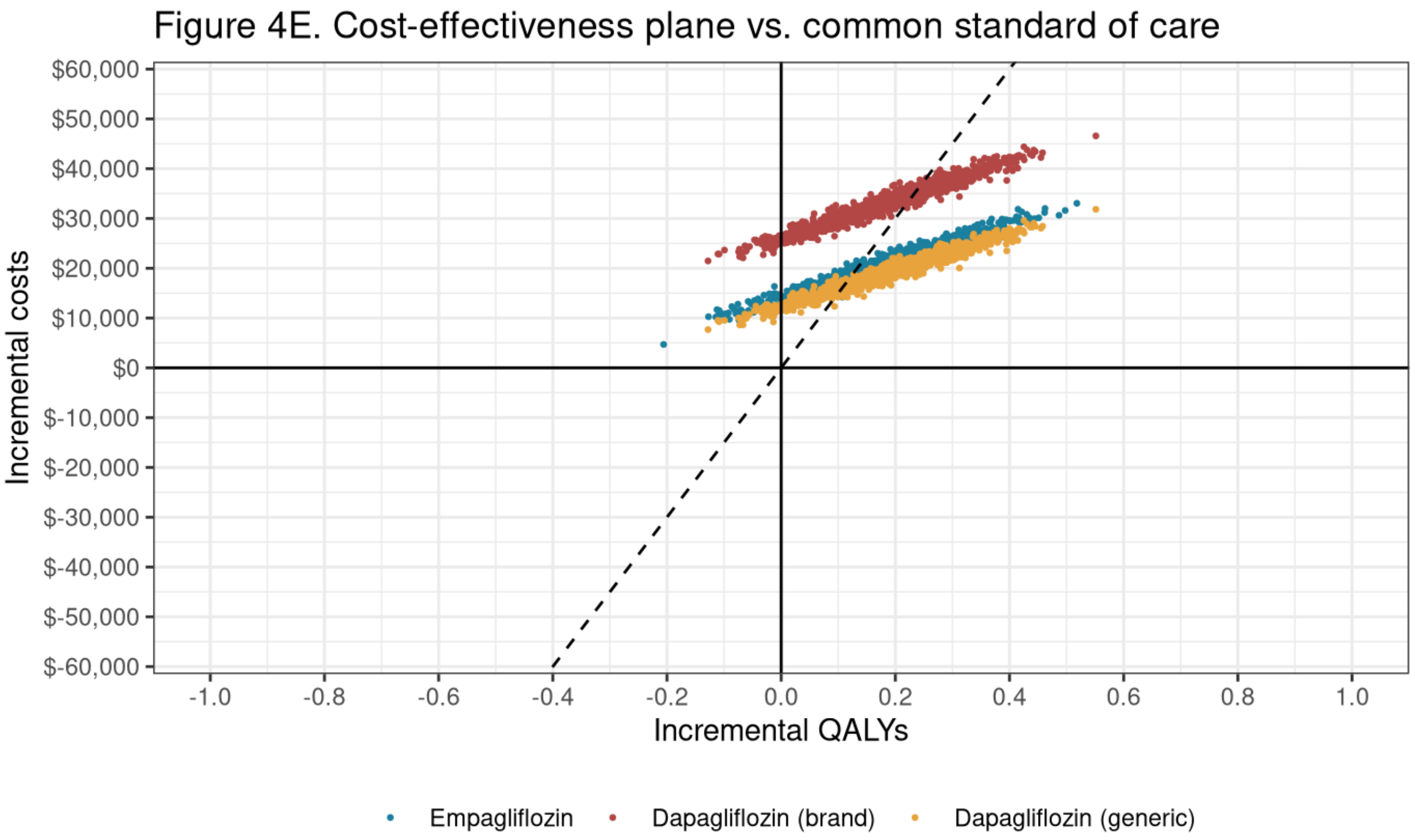


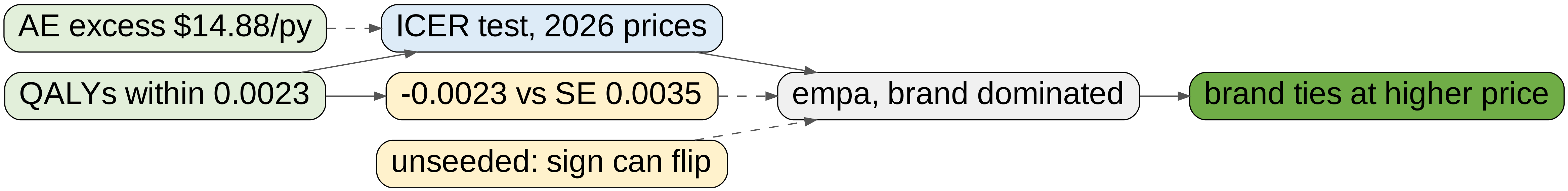
Figure 4F: Dominance and sequential ICERs

Strategy	Mean QALYs	Mean cost	Seq. ICER	Status
Standard of care	5.158	$167,513	–	on frontier
Dapagliflozin (generic)	5.343	$186,492	$102,367	on frontier
Empagliflozin	5.341	$188,441	–	dominated
Dapagliflozin (brand)	5.343	$200,672	–	dominated

Classification is **identical** with adverse-event costs removed.

How much weight the frontier will bear

All three within **0.0023 QALYs**: no measurable QALY advantage.

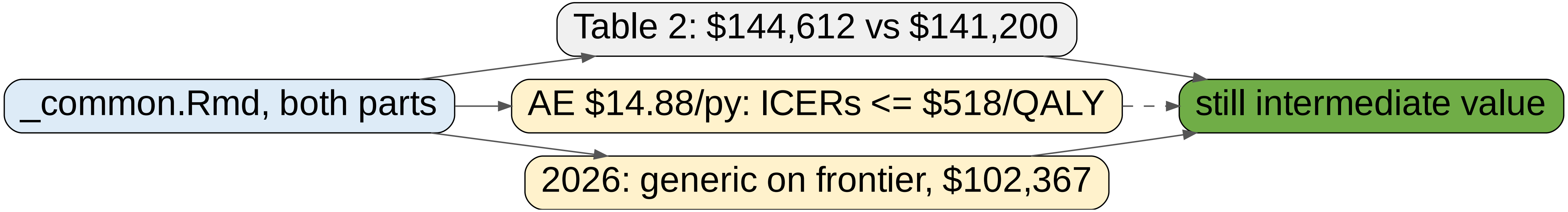


2026 prices · the three-drug numbers · seeding.

Conclusions

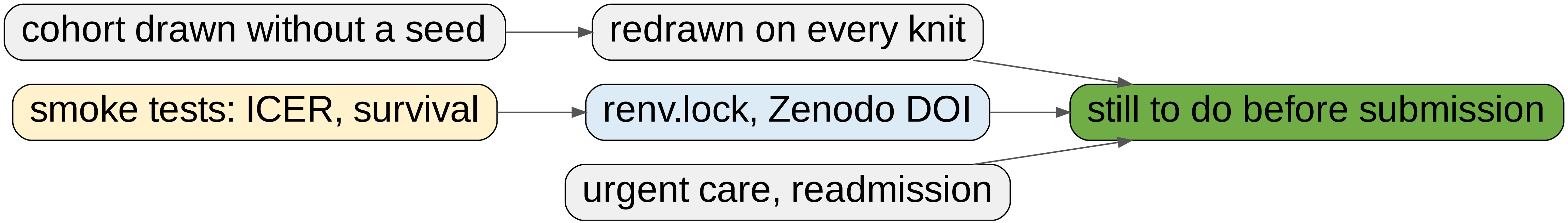
What the two documents establish

The model reproduces; price, not evidence, reorders the drugs.



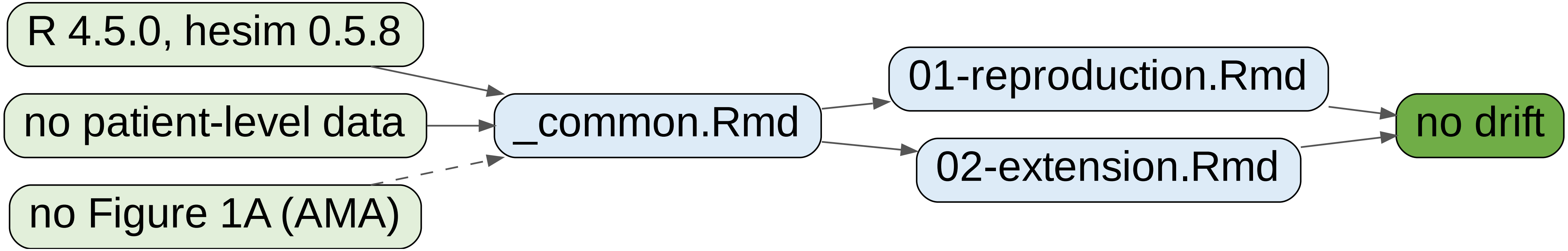
the two documents · Table 2 · the three-drug results.

Limitations and next steps



structural gaps · adverse events · the survival check.

Reproducibility



References

1. Cohen LP, Isaza N, Hernandez I, et al. Cost-effectiveness of sodium-glucose cotransporter-2 inhibitors for the treatment of heart failure with preserved ejection fraction. *JAMA Cardiol.* 2023;8(5):419-428. doi:10.1001/jamacardio.2023.0077
2. Incerti D, Jansen JP. hesim: health economic simulation modeling and decision analysis. *arXiv.* 2021;2102.09437. https://arxiv.org/abs/2102.09437
3. Anker SD, Butler J, Filippatos G, et al. Empagliflozin in heart failure with a preserved ejection fraction. *N Engl J Med.* 2021;385(16):1451-1461. doi:10.1056/NEJMoa2107038
4. Solomon SD, McMurray JJV, Claggett B, et al. Dapagliflozin in heart failure with mildly reduced or preserved ejection fraction. *N Engl J Med.* 2022;387(12):1089-1098. doi:10.1056/NEJMoa2206286
5. Vaduganathan M, Docherty KF, Claggett BL, et al. SGLT-2 inhibitors in patients with heart failure: a comprehensive meta-analysis of 5 randomized controlled trials. *Lancet.* 2022;400(10354):757-767. doi:10.1016/S0140-6736(22)01429-5
6. US Department of Veterans Affairs. VA Federal Supply Schedule Service. Accessed January 28, 2022. https://www.fss.va.gov/
7. Arias E, Xu J. US life tables, 2018. *Natl Vital Stat Rep.* 2020;69(12):1-45.
8. US Centers for Disease Control and Prevention. CDC WONDER. Accessed September 18, 2021. https://wonder.cdc.gov
9. Cheng RK, Cox M, Neely ML, et al. Outcomes in patients with heart failure with preserved, borderline, and reduced ejection fraction in the Medicare population. *Am Heart J.* 2014;168(5):721-730. doi:10.1016/j.ahj.2014.07.008
10. Arora S, Lahewala S, Hassan Virk HU, et al. Etiologies, trends, and predictors of 30-day readmissions in patients with diastolic heart failure. *Am J Cardiol.* 2017;120(4):616-624. doi:10.1016/j.amjcard.2017.05.028
11. Goyal P, Loop M, Chen L, et al. Causes and temporal patterns of 30-day readmission among older adults hospitalized with heart failure with preserved or reduced ejection fraction. *J Am Heart Assoc.* 2018;7(9):e007785. doi:10.1161/JAHA.117.007785
12. Shah KS, Xu H, Matsouaka RA, et al. Heart failure with preserved, borderline, and reduced ejection fraction: 5-year outcomes. *J Am Coll Cardiol.* 2017;70(20):2476-2486. doi:10.1016/j.jacc.2017.08.074
13. Goyal P, Almarzooq ZI, Horn EM, et al. Characteristics of hospitalizations for heart failure with preserved ejection fraction. *Am J Med.* 2016;129(6):635.e15-635.e26. doi:10.1016/j.amjmed.2016.02.007
14. Chamberlain AM, Dunlay SM, Gerber Y, et al. Burden and timing of hospitalizations in heart failure: a community study. *Mayo Clin Proc.* 2017;92(2):184-192. doi:10.1016/j.mayocp.2016.11.009
15. Sanders GD, Neumann PJ, Basu A, et al. Recommendations for conduct, methodological practices, and reporting of cost-effectiveness analyses: second panel on cost-effectiveness in health and medicine. *JAMA.* 2016;316(10):1093-1103. doi:10.1001/jama.2016.12195
16. Butler J, Filippatos G, Jamal Siddiqi T, et al. Empagliflozin, health status, and quality of life in patients with heart failure and preserved ejection fraction: the EMPEROR-Preserved trial. *Circulation.* 2022;145(3):184-193. doi:10.1161/CIRCULATIONAHA.121.057812
17. Thomas M, Jones PG, Cohen DJ, et al. Predicting the EQ-5D utilities from the Kansas City Cardiomyopathy Questionnaire in patients with heart failure. *Eur Heart J Qual Care Clin Outcomes.* 2021;7(4):388-396. doi:10.1093/ehjqcco/qcab014
18. Isaza N, Calvachi P, Raber I, et al. Cost-effectiveness of dapagliflozin for the treatment of heart failure with reduced ejection fraction. *JAMA Netw Open.* 2021;4(7):e2114501. doi:10.1001/jamanetworkopen.2021.14501
19. Bryant KB, Moran AE, Kazi DS, et al. Cost-effectiveness of hypertension treatment by pharmacists in Black barbershops. *Circulation.* 2021;143(24):2384-2394. doi:10.1161/CIRCULATIONAHA.120.051683
20. Kazi DS, Bellows BK, Baron SJ, et al. Cost-effectiveness of tafamidis therapy for transthyretin amyloid cardiomyopathy. *Circulation.* 2020;141(15):1214-1224. doi:10.1161/CIRCULATIONAHA.119.045093
21. HCUPnet. Healthcare Cost and Utilization Project (HCUP): introduction to the HCUP Nationwide Inpatient Sample (NIS) 2018. Published 2020. Accessed September 10, 2021. https://www.hcup-us.ahrq.gov/db/nation/nis/NISIntroduction2018.pdf
22. Peterson C, Xu L, Florence C, Grosse SD, Annest JL. Professional fee ratios for US hospital discharge data. *Med Care.* 2015;53(10):840-849. doi:10.1097/MLR.0000000000000410
23. Sullivan PW, Ghushchyan V. Preference-based EQ-5D index scores for chronic conditions in the US. *Med Decis Making.* 2006;26(4):410-420. doi:10.1177/0272989X06290495
24. Jaagosild P, Dawson NV, Thomas C, et al. Outcomes of acute exacerbation of severe congestive heart failure: quality of life, resource use, and survival. *Arch Intern Med.* 1998;158(10):1081-1089. doi:10.1001/archinte.158.10.1081
25. Packer M, Butler J, Zannad F, et al. Effect of empagliflozin on worsening heart failure events in patients with heart failure and preserved ejection fraction: EMPEROR-Preserved trial. *Circulation.* 2021;144(16):1284-1294. doi:10.1161/CIRCULATIONAHA.121.056824
26. Ferreira JP, Zannad F, Butler J, et al. Genitourinary tract infections and SGLT2 inhibitors in heart failure: an EMPEROR-Pooled analysis. *Eur J Heart Fail.* Published online February 28, 2026. doi:10.1093/ejhf/xuag054
27. US Department of Veterans Affairs, Office of Procurement, Acquisition and Logistics, National Acquisition Center. Pharmaceutical pricing data as of March 1, 2026: Federal Supply Schedule and National Contracts prices for all National Acquisition Center programs. Accessed July 29, 2026. https://department.va.gov/procurement-acquisition-and-logistics/wp-content/uploads/sites/56/2026/03/TPharmPrices2026-03-01.xlsx
28. Rane A, Nechi RN, Imam M, et al. Comparative value of dapagliflozin vs empagliflozin in patients with heart failure and preserved ejection fraction: a cost-effectiveness analysis. *J Manag Care Spec Pharm.* 2023;29(9):1045-1053. doi:10.18553/jmcp.2023.29.9.1045